

CivicRadar: Hyperlocal Civic Issue Reporting and Resolution Platform

Hackathon: Vibe2Ship

Project: CivicRadar

Selected Problem Statement: Community Hero - Hyperlocal Problem Solver

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Deployed App: <https://civicradar-378091373108.asia-south1.run.app>

GitHub Repository: <https://github.com/official-vanshaj-garg/civicradar-vibe2ship>

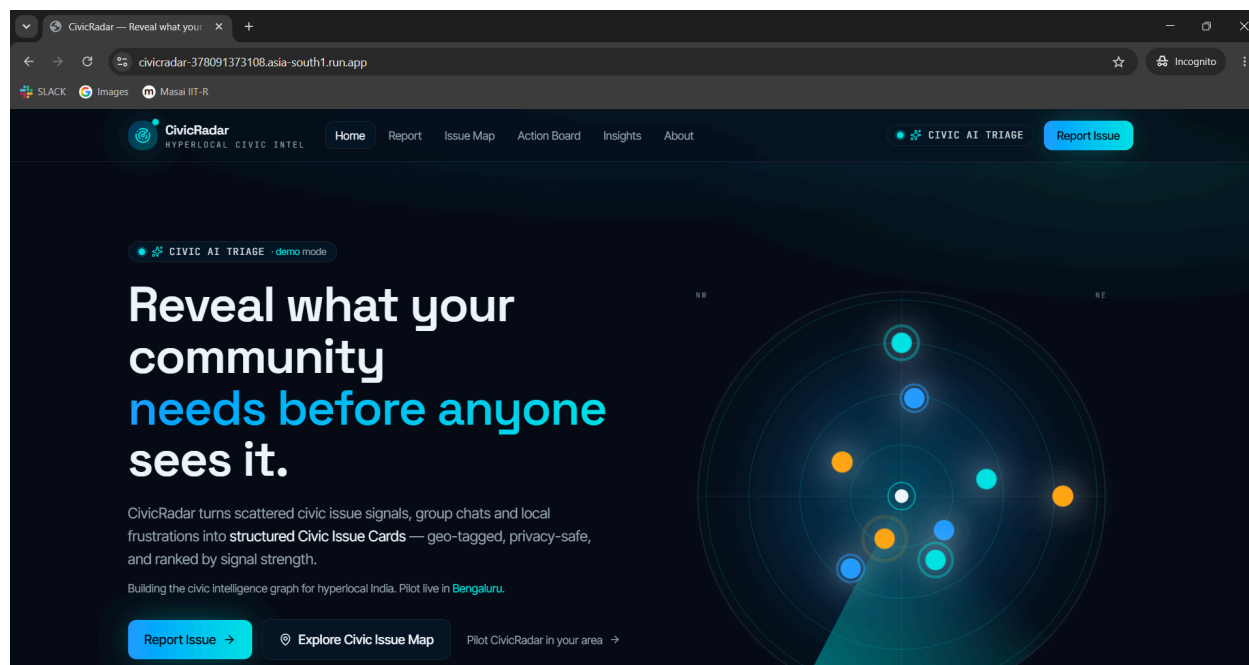


Figure 1: CivicRadar home page showing the hyperlocal civic reporting mission.

1. Problem Statement Selected

Selected: Community Hero - Hyperlocal Problem Solver.

Communities deal with everyday local issues such as potholes, broken streetlights, garbage accumulation, water leakage, and unsafe roads. Most get reported informally through group chats, calls, or scattered local platforms, which makes them hard to validate, prioritize, or track over time.

The problem statement calls for a platform that lets citizens identify, report, validate, track, and resolve community issues through collaboration, data, and intelligent automation, while encouraging transparency and accountability.

In practice, this breaks down into four gaps CivicRadar tries to close:

- **Reporting gap:** citizens notice issues but lack a structured way to report them.
- **Validation gap:** communities cannot easily confirm whether an issue affects one person or many.
- **Accountability gap:** reports rarely make clear who should respond or what should happen next.
- **Tracking gap:** citizens usually cannot see a transparent path from report to resolution.

2. Solution Overview

CivicRadar is a hyperlocal civic issue reporting and tracking platform built for the Community Hero problem statement.

It lets citizens report local issues, get a structured civic action packet for each one, add evidence metadata, verify issues reported by others, follow an issue through its lifecycle, and see civic activity on a dashboard and map.

The current build centers on a Bengaluru pilot and demonstrates the full loop:

Report → Triage → Validate → Route → Track → Follow up / Resolve in demo

CivicRadar does not claim official government dispatch or authority integration. It is a working prototype that shows how a community-first civic reporting loop can function in a safe, transparent way.

3. Key Features

CivicRadar's feature set covers the full reporting and accountability loop, from first report to follow-up.

Civic Issue Reporting

Citizens submit an issue by describing it, picking a category and location zone, and adding context. The flow covers common cases such as potholes, waste accumulation, broken streetlights, water leakage, unsafe public areas, and road or infrastructure problems. Once submitted, the issue appears on the dashboard and the civic map.

Civic Triage and Action Packet

Every report runs through a deterministic triage system that turns it into a structured action packet instead of leaving it as a plain complaint. The packet includes:

- Issue category and urgency level
- Affected group and civic priority score
- Community signal strength
- Responsible stakeholder
- Suggested next action
- Why the issue matters

This is meant to help RWAs, NGOs, campus teams, and local volunteers understand what an issue is and who is best placed to act on it. CivicRadar treats stakeholder routing as a recommendation, not a claim that any official dispatch happens.

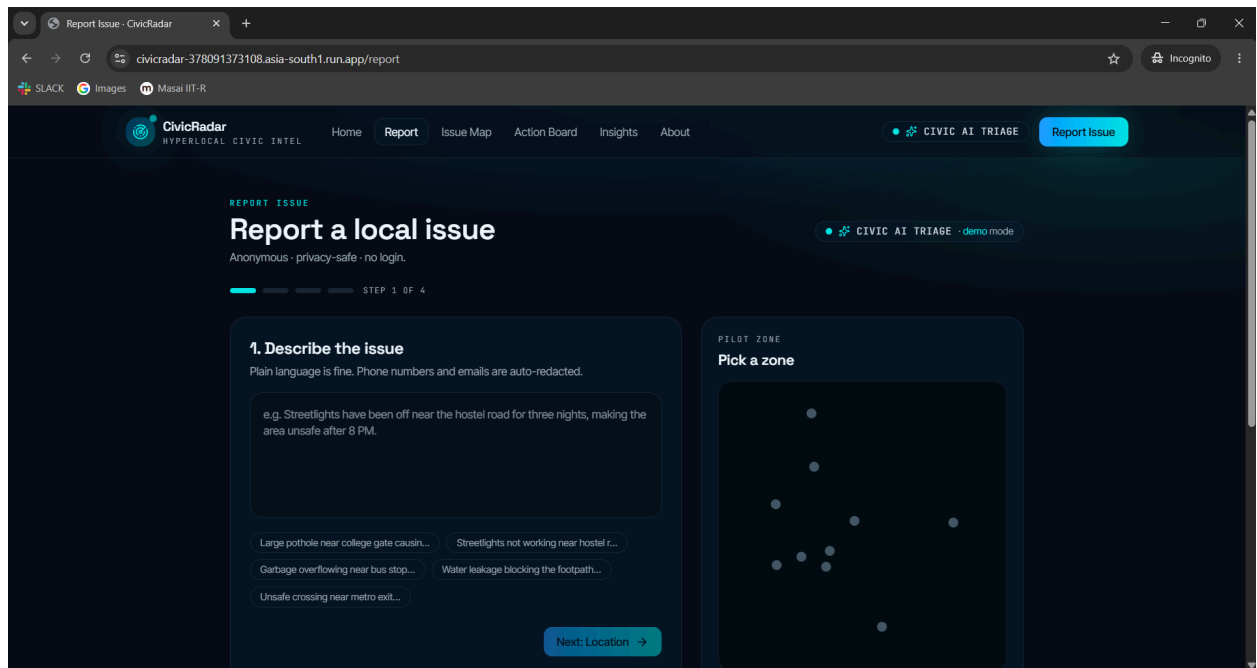
The image is a screenshot of a web browser showing the 'Report a local issue' form on the CivicRadar website. The browser's address bar shows the URL 'civicsradar-378091373108.asia-south1.run.app/report'. The website's navigation bar includes links for Home, Report, Issue Map, Action Board, Insights, and About. A 'Report Issue' button is visible in the top right. The main heading is 'Report a local issue' with a subtext 'Anonymous · privacy-safe · no login.' and a progress indicator 'STEP 1 OF 4'. The first step, '1. Describe the issue', includes a text area with a placeholder example: 'e.g. Streetlights have been off near the hostel road for three nights, making the area unsafe after 8 PM.' Below the text area are several radio button options: 'Large pothole near college gate causin...', 'Streetlights not working near hostel r...', 'Garbage overflowing near bus stop...', 'Water leakage blocking the footpath...', and 'Unsafe crossing near metro exit...'. A 'Next: Location' button is at the bottom right of this section. To the right of the description section is a 'PILOT ZONE' section titled 'Pick a zone' which contains a map with several blue dots representing different zones.

Figure 2: Citizen report flow with evidence metadata and optional approximate location assist.

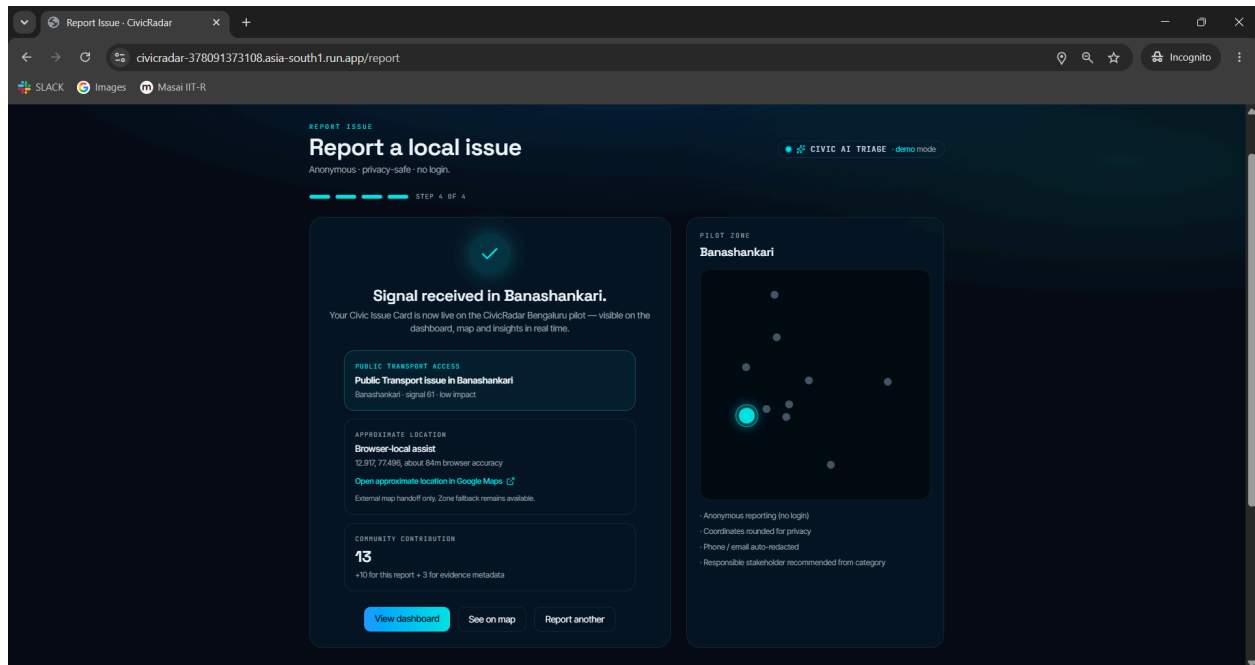


Figure 3: Generated civic action packet with priority, stakeholder routing, and suggested next action.

Evidence Metadata

Reports can note the kind of supporting evidence available: no evidence, photo metadata, video metadata, or a witness note. The current prototype does not upload actual files; it stores the evidence type as metadata only. This keeps the prototype simple and safe while still showing how evidence-backed reporting would work in a full system. The UI states clearly that evidence is metadata only.

Community Verification

From the map or issue drawer, other users can verify a reported issue. The system supports "verify this issue," shows "verified by you," and keeps a local verification count, limited to one verification per issue per browser. This supports the validation part of the problem statement by showing how multiple citizens can strengthen a signal instead of relying on a single report. Verification is currently browser-local, stored with localStorage.

Civic Lifecycle Timeline

Each issue carries a visible lifecycle: Reported, AI triaged, Community verified, Routed, and Follow-up needed / Resolved in demo. This gives users a sense of progress instead of a static report. The resolved state is clearly marked as demo-local; CivicRadar does not claim real government resolution.

Civic Contribution Score

A simple, non-gamified contribution score encourages participation: +10 for submitting a report, +5 for verifying an issue for the first time, and +3 for adding evidence metadata.

Civic Dashboard

The dashboard summarizes civic activity: top issues needing action, civic priority signals, verified issue count, contribution score, community signal strength, and accountability insights, so users can quickly see what needs attention.

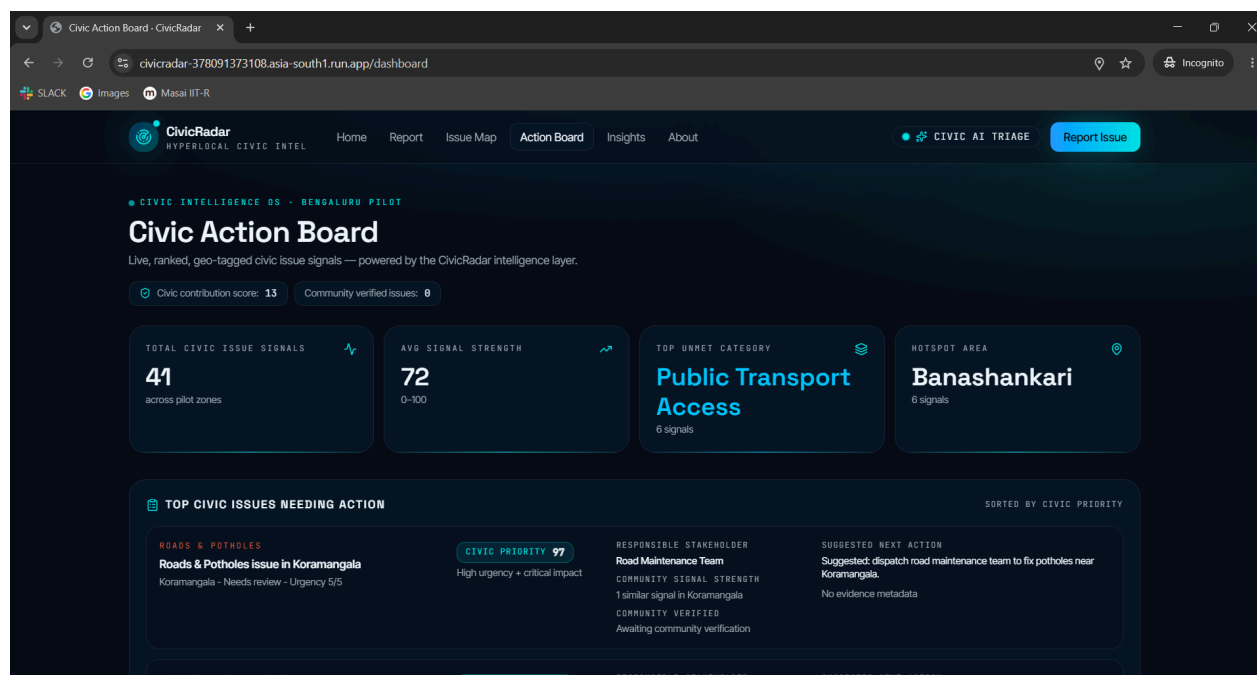


Figure 4: Civic dashboard showing contribution score, verified signals, and priority issues.

Hyperlocal Civic Map

A dark-themed civic map shows issue locations across Bengaluru pilot zones. Clicking a marker opens a drawer with issue details, the action packet, evidence metadata, verification status, lifecycle timeline, stakeholder routing, and a Google Maps handoff link when coordinates are available.

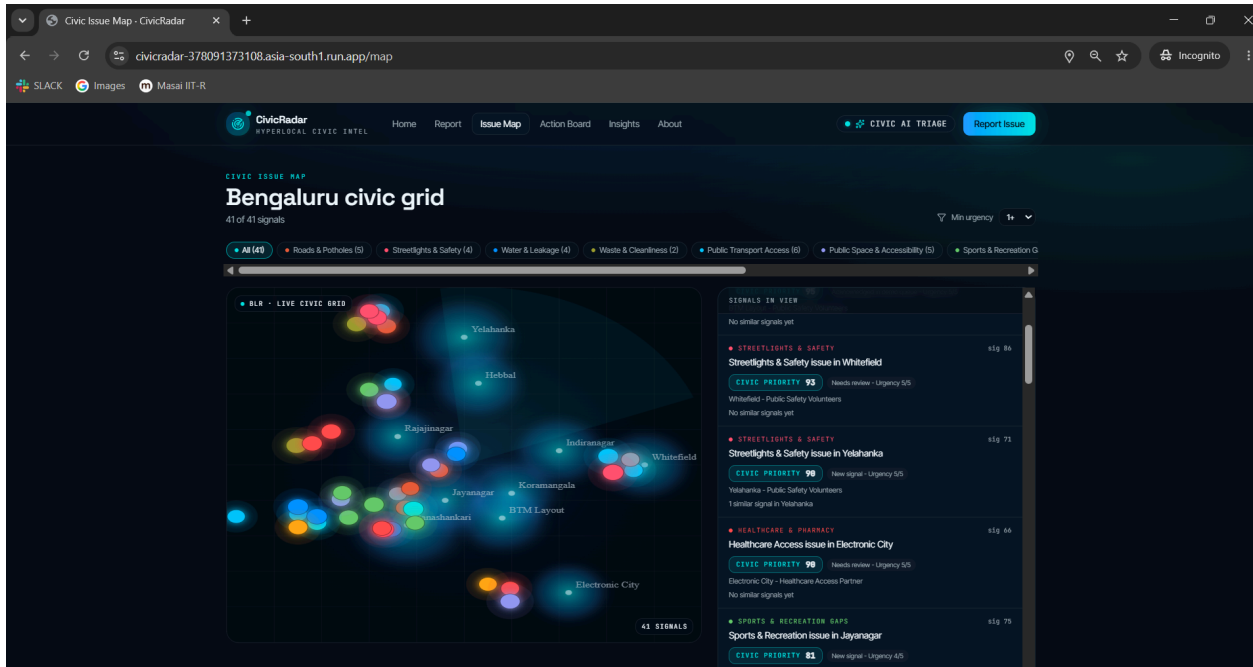


Figure 5: Map issue drawer with community verification, lifecycle timeline, and Google Maps handoff.

Location Assist

The report flow includes an optional "Use approximate current location" button. Location is never requested automatically, and if permission is denied or unavailable, the app falls back to zone selection. Report submission is never blocked by a geolocation failure, and the app does not claim exact GPS verification.

4. Technologies Used

Frontend: React 19, TypeScript, Vite, TanStack Router / TanStack Start, Tailwind CSS, shadcn/ui, Radix UI, Lucide icons

Tooling: Bun, ESLint, TypeScript type checking, Git and GitHub

Storage: Browser localStorage for prototype persistence, covering user-created reports, verification counts and verified issue IDs, demo resolution state, contribution score, and approximate location metadata.

This lets the prototype demonstrate the full civic loop without a backend database or collecting personal data beyond what a user explicitly submits.

5. Google Technologies Utilized

Google Cloud Run

CivicRadar is deployed publicly on Google Cloud Run, which hosts the application and keeps it accessible for the evaluation period.

Live app: <https://civicradar-378091373108.asia-south1.run.app>

Google Cloud Project

The app is deployed under a Google Cloud project and served as a Cloud Run service.

Google Maps External Handoff

For issues with approximate coordinates, CivicRadar links out to Google Maps using a standard search URL, for example:

<https://www.google.com/maps/search/?api=1&query=lat,lng>

This does not use the Google Maps JavaScript API, does not embed a map script, and does not use a Maps API key. It gives judges and users a simple way to check approximate issue locations without adding API-key risk to the deployment.

6. Agentic and Intelligent Workflow

CivicRadar's "intelligence" is a deterministic civic triage system, not a live AI model. It takes a plain citizen report and turns it into a structured civic action packet: category, urgency, civic priority score, community signal strength, responsible stakeholder, suggested next action, and a short explanation of why the issue matters, alongside lifecycle and verification status.

This is what makes the platform action-oriented rather than just a reporting form. It is deterministic by design, for reliability and demo safety. The Production Roadmap section below covers how this could be extended with server-side AI-based triage in a real deployment.

7. Product Experience and Design

CivicRadar uses a dark, radar-inspired visual style built around four main surfaces:

- **Home:** explains the product and its civic mission
- **Report:** lets citizens submit structured issue reports
- **Dashboard:** shows civic signals, contribution score, and priority issues
- **Map:** shows issue locations, verification, and lifecycle details

The design aims for a local civic command center feel, with clarity and accountability prioritized over claiming any official authority.

8. Prototype Scope and Honesty

CivicRadar is a working hackathon prototype, and it is built to be upfront about what that means:

- Reports are saved in the browser using localStorage and update the dashboard and map in that same browser.
- Verification is local to the same browser and device.
- Evidence is metadata only; no files are uploaded.
- Approximate location is optional and only used when the user triggers it.
- No official government dispatch is performed, and no government integration is claimed.
- There is no backend database in the current version.

This scope was a deliberate choice. localStorage keeps the prototype safe, fast to build, and easy to evaluate without standing up a backend within the hackathon timeline, while still proving out the full report-to-resolution loop end to end.

9. Production Roadmap

A production version of CivicRadar would build on this prototype with:

- Cloud Firestore, or a Cloud Run API backed by Firestore, for shared civic reports across users
- Cloud Storage for optional image or video evidence uploads
- Server-side Gemini-based triage for richer civic issue understanding
- Rate limiting and abuse prevention
- A moderator dashboard for RWAs, ward teams, NGOs, and campus groups
- Notification and follow-up workflows
- Verified stakeholder routing
- Public issue status pages
- Analytics for civic hotspots and recurring problems

The prototype focuses on proving the core loop first. This roadmap is the path to a production-grade civic platform.

10. Recommended Demo Flow for Judges

1. Open the live app.
2. Go to the Report page and submit a civic issue with evidence metadata.
3. Optionally use the approximate location button.
4. Review the generated civic action packet.

5. Open the Dashboard to see the contribution score and civic signals.
6. Open the Map, click an issue marker, and review the drawer.
7. Verify an issue and check the lifecycle timeline.
8. Open the approximate location in Google Maps, if available.

11. Evaluation Alignment

A quick map from CivicRadar's features to the evaluation criteria:

- **Problem Solving & Impact:** CivicRadar turns fragmented, informal civic reporting into a structured flow for reporting, validating, tracking, and following up on local issues.
- **Agentic Depth:** Every report is converted into a civic action packet with priority, stakeholder routing, a suggested next action, and a lifecycle status, instead of staying a static complaint.
- **Innovation & Creativity:** The product combines reporting, community verification, accountability packets, contribution scoring, lifecycle tracking, and a civic map and dashboard into a single workflow.
- **Usage of Google Technologies:** The app is deployed on Google Cloud Run and uses Google Maps external handoff links for approximate issue locations.
- **Product Experience & Design:** A dark, radar-inspired interface ties report, dashboard, map, and issue-detail flows into a civic command center feel.
- **Technical Implementation:** Built with React, TypeScript, Vite, TanStack, structured domain models, and localStorage persistence, deployed on Cloud Run.
- **Completeness & Usability:** The deployed app supports a full demo loop, from report creation through triage, verification, dashboard visibility, map tracking, and demo-local resolution.

12. Final Submission Summary

CivicRadar is a hyperlocal civic issue platform built for the Community Hero - Hyperlocal Problem Solver challenge.

It lets citizens report local issues, get a structured civic action packet, add evidence metadata, validate issues through community verification, track lifecycle status, and view civic signals on a dashboard and map.

The app is deployed on Google Cloud Run and demonstrates a complete civic reporting and accountability loop, while staying honest about what is and is not real in the current prototype.

13. Short Description for Platform Field

CivicRadar is a hyperlocal civic issue reporting and accountability platform for the Community Hero problem statement. Citizens report local issues, get a structured civic action packet, add

evidence metadata, verify issues, track lifecycle status, and view civic signals on a dashboard and map. The app is deployed on Google Cloud Run and uses browser-local persistence, community verification, contribution scoring, optional approximate geolocation, and an external Google Maps handoff. It is an honest prototype: no official government dispatch, no file uploads, and no backend database in the current version.